Running every street in Paris with Python and PostGIS 🗼

PyCon US 2026

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Outline

- Motivation
- GIS and related concepts
- OpenStreetMap and street networks
- Storing data in Postgres with PostGIS
- Map matching runs and visualizing progress

vinayak.io/code

Camelot

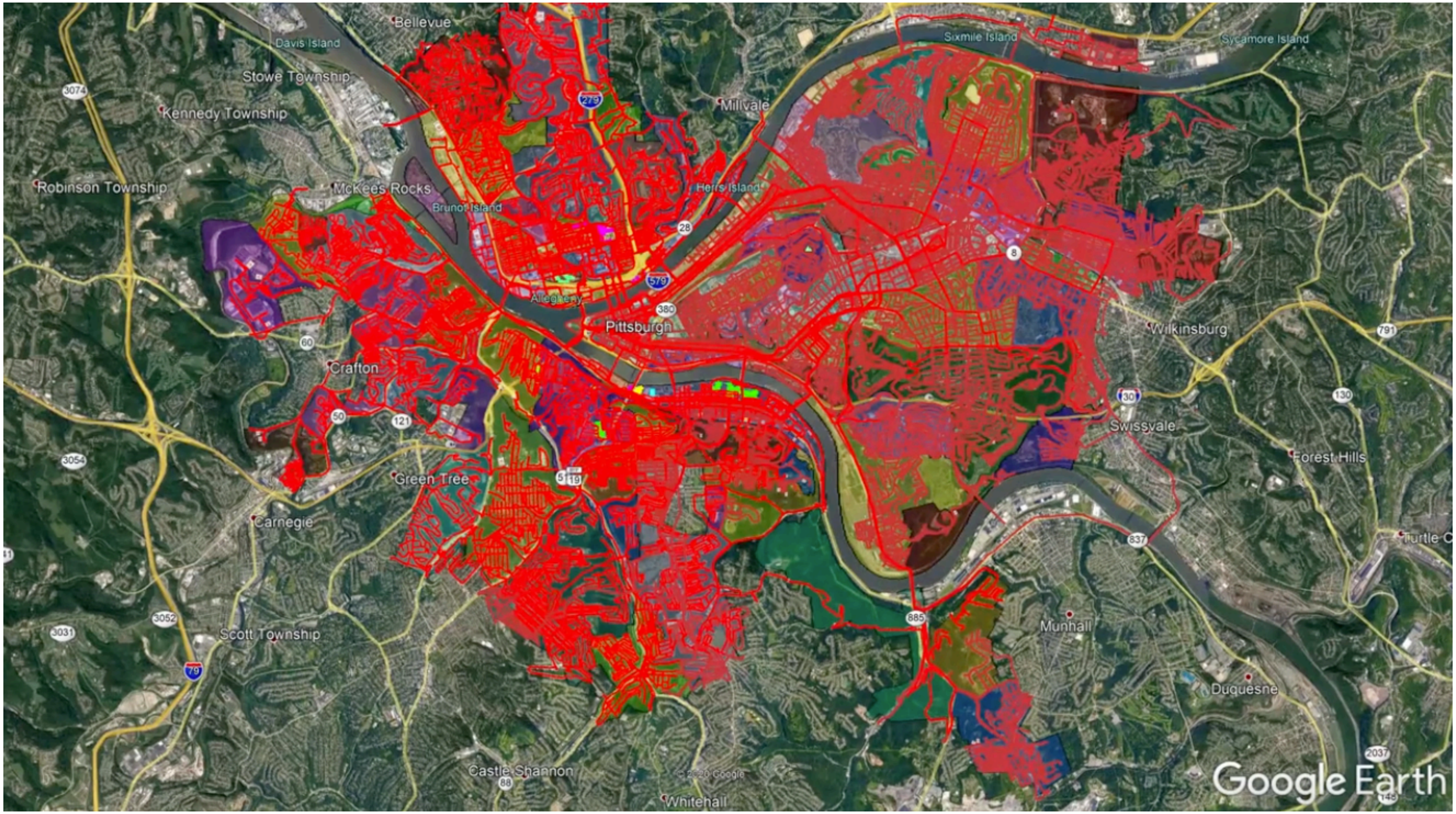
Present

Fleur

recurse.com

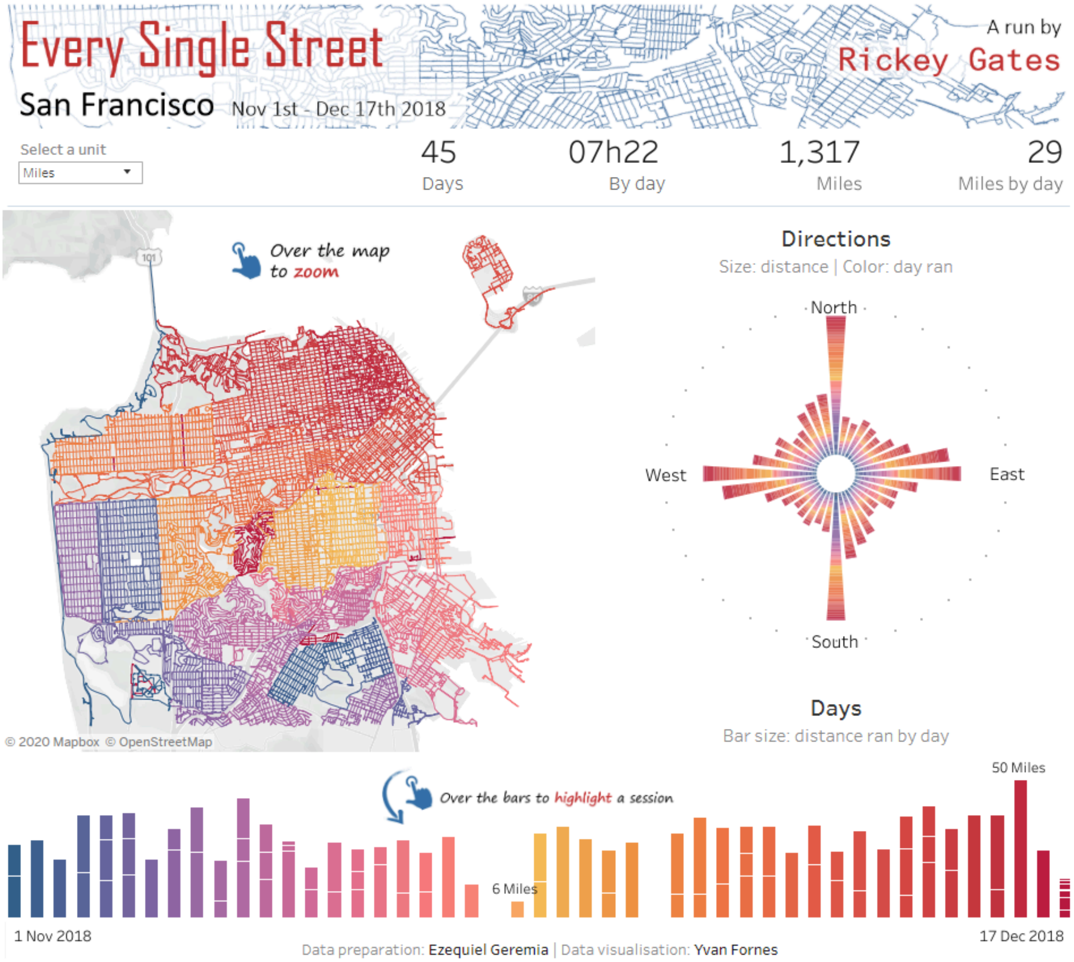
Running every street in Paris

Tom Murphy's PAC TOM project



How I ran the length of every street in Pittsburgh: PAC TOM

Rickey Gates' Every Single Street project



https://www.rickeygates.com/everysinglestreet

"Could I do it in Paris?!"

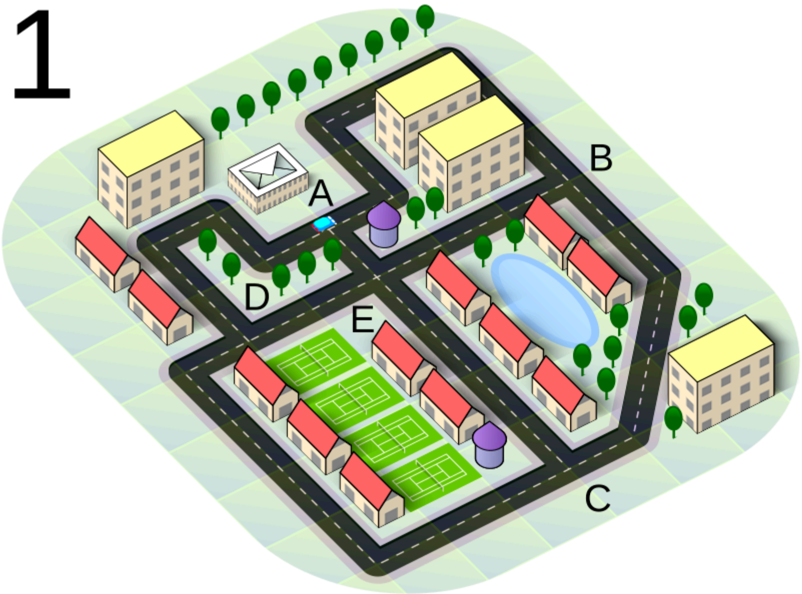
Approach

- Fetch OSM data
- Compute routes
- Run 🏃
- Map match
- Visualize progress

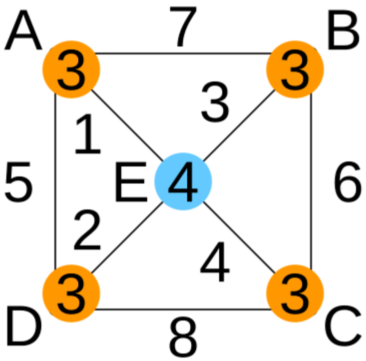
OpenStreetMap

Chinese postman problem

1



2



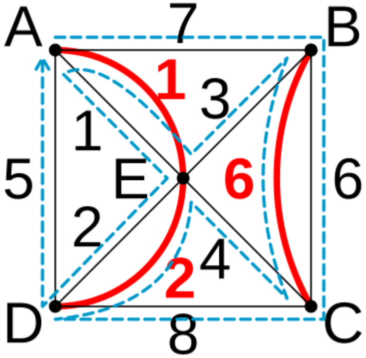
3

$\mathrm{A(E)B} + \mathrm{C(E)D} = 4 + 6 = 10$

$\mathrm{A(E)C} + \mathrm{B(E)D} = 5 + 5 = 10$

$\mathrm{A(E)D} + \mathrm{BC} \quad = 3 + 6 = 9$ ✔

4



$$L = 7+6+8+\mathbf{2}+4+\mathbf{6}+3+\mathbf{1}+1+2+5 = 45$$

Approach (revised)

- Fetch OSM data
- ~~Compute routes~~ Draw routes
- Run 🏃
- Map match
- Visualize progress

Progress

9:14 PM on Wednesday, April 15, 2026 · Paris

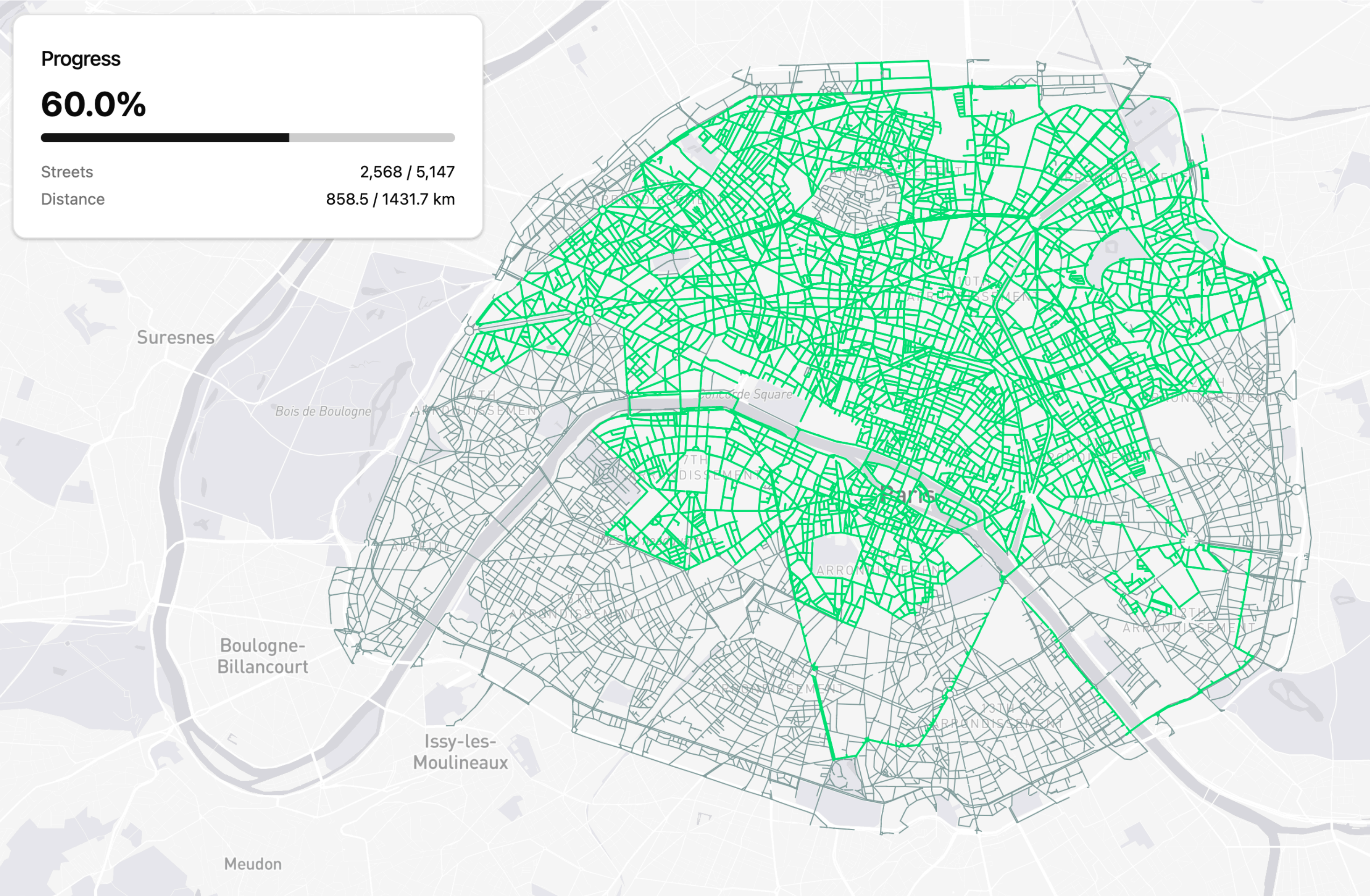
Night Run

- Streets completed: 2,566 / 5,147
- Progress: 60% (📈 +1%)
- City: Paris

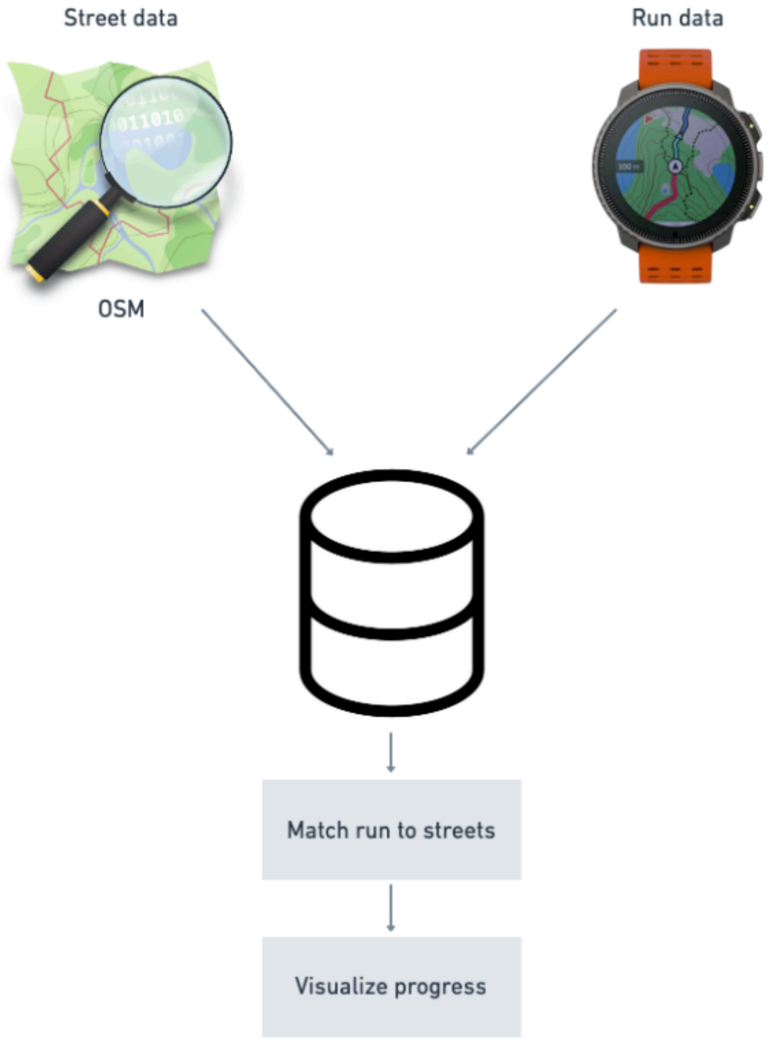
🗺️ Check out my progress at https://app.everystreet.run/vinayak/paris 🏃

•••

Progress

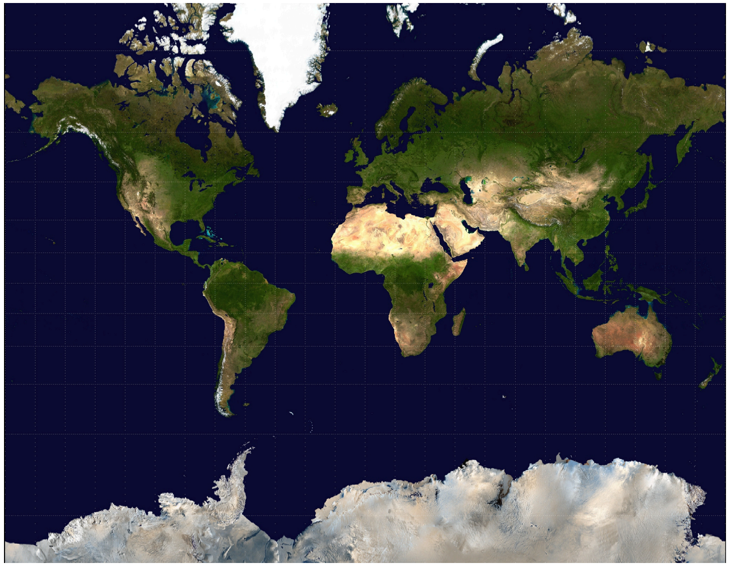

Progress
60.0%
Streets 2,568 / 5,147
Distance 858.5 / 1431.7 km
Suresnes
Bois de Boulogne
Boulogne-Billancourt
Issy-les-Moulineaux
Meudon
Paris

GIS

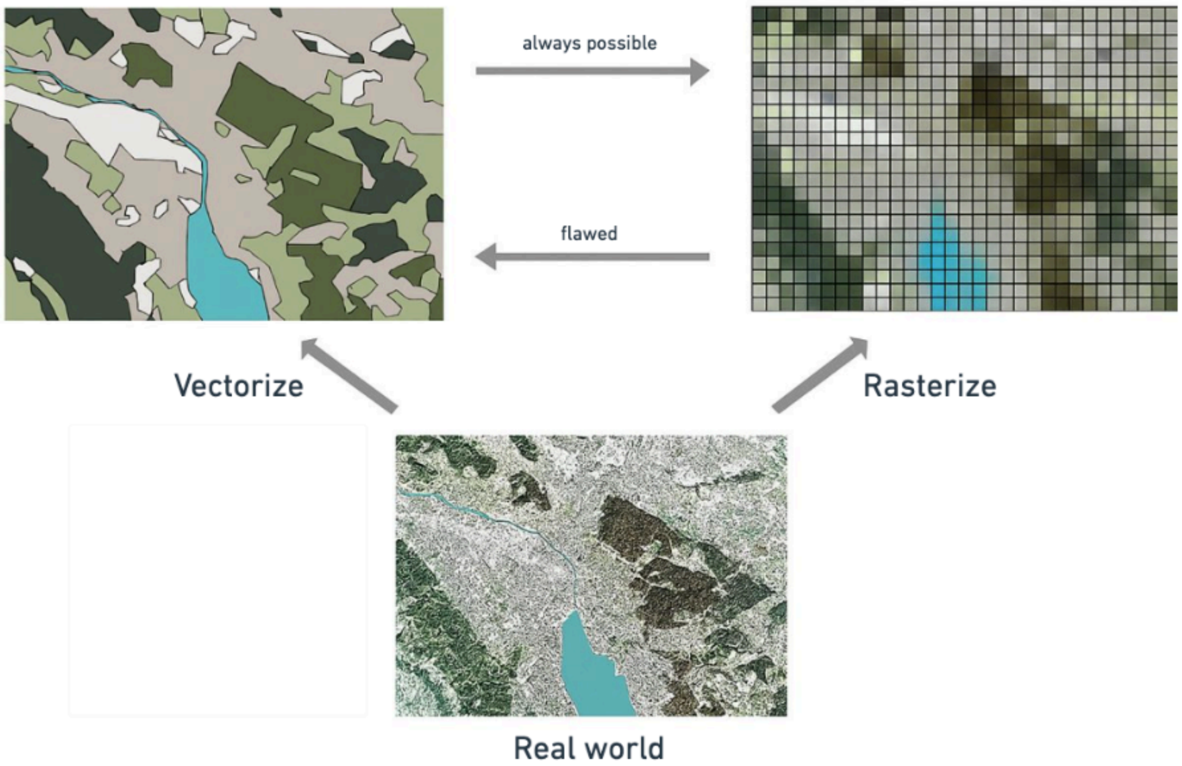

Street data
Run data
OSM
Match run to streets
Visualize progress

Coordinate systems

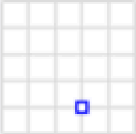
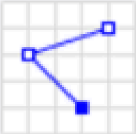
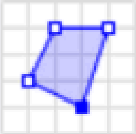
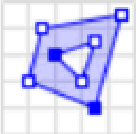
- Geographic coordinate systems (WGS 84, ...)
- Projected coordinate systems (Mercator, ...)



Vector and raster data



Geometries

Type	Examples	
Point		`POINT (30 10)`
LineString		`LINESTRING (30 10, 10 30, 40 40)`
Polygon		`POLYGON ((30 10, 40 40, 20 40, 10 20, 30 10))`
		`POLYGON ((35 10, 45 45, 15 40, 10 20, 35 10), (20 30, 35 35, 30 20, 20 30))`

GPX

```xml
<!--?xml version="1.0" encoding="UTF-8"?-->
<gpx version="1.1" creator="Strava">
  <metadata>
    <name>Morning Run</name>
    <time>2025-07-10T22:00:00Z</time>
  </metadata>
  <trk>
    <name>Sample Track</name>
    <trkseg>
      <trkpt lat="48.8566" lon="2.3522">
        <ele>35.0</ele>
        <time>2025-07-10T22:00:00Z</time>
      </trkpt>
      <trkpt lat="48.8570" lon="2.3530">
```

Postgres and PostGIS

```sql
-- Enable PostGIS extension
CREATE EXTENSION postgis;
```

Let's look at some code


RUE
MILTON
é • Fraternité
HOTEL
INTERNATIONAL VISUAL THEATRE
IVT

Future work

- Viterbi algorithm
- ~~Draw routes~~ Compute routes
- Full coverage this year

Questions

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